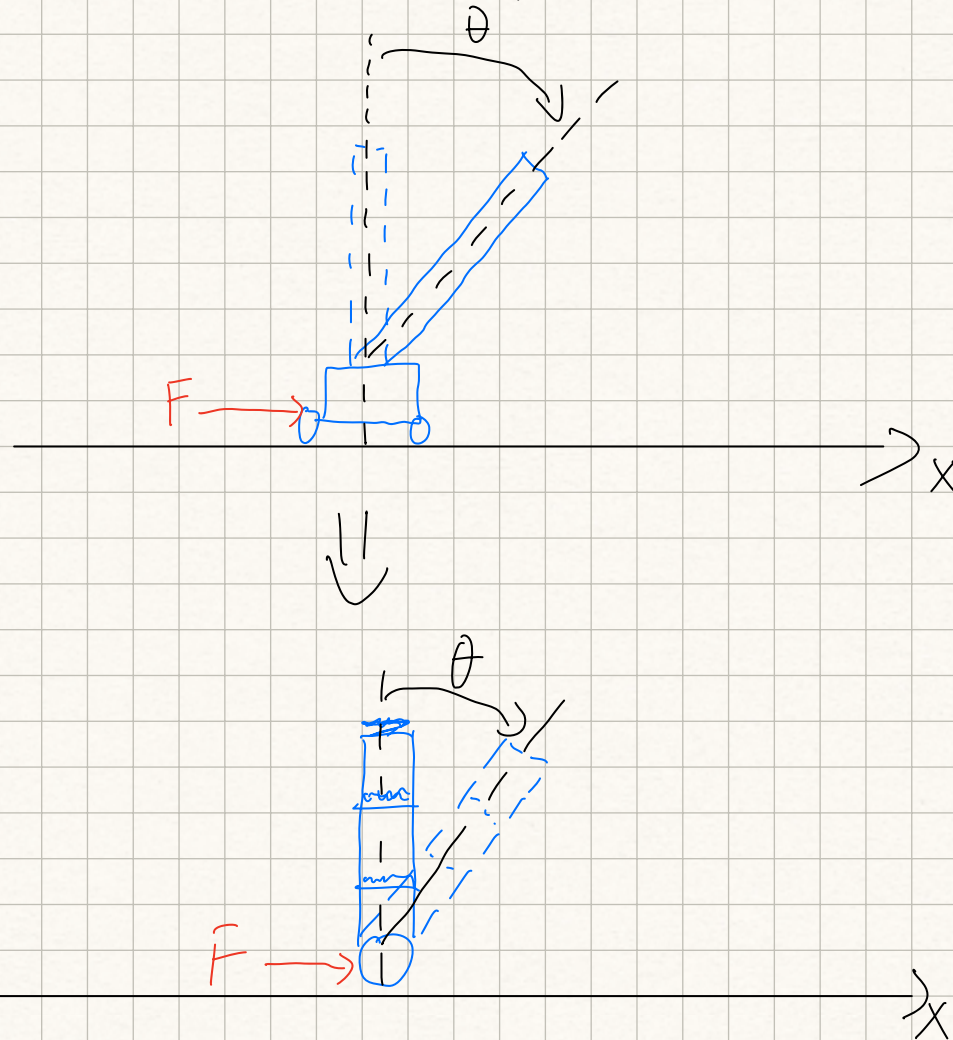


System Model

Self-Balancing Robot

Modeled as an inverted pendulum on a cart:



Goal:

Keep the robot upright ($\theta = 0$) by applying voltages to the two motors controlling the wheels.

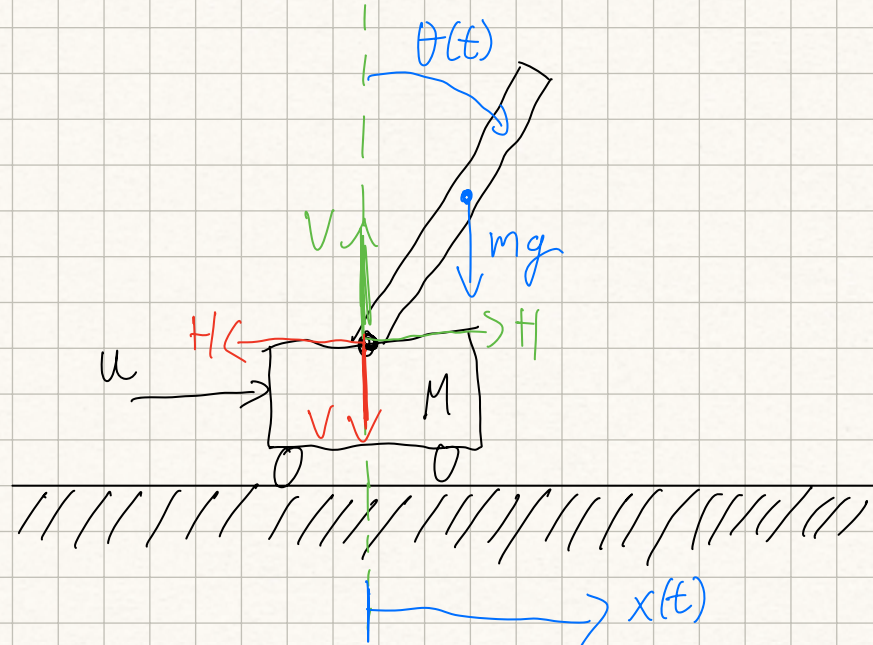
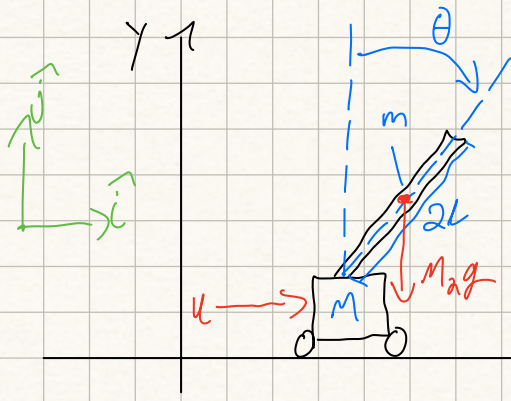
State variables:

θ : tilt angle of body

$\dot{\theta}$: angular velocity

x : horizontal position of robot base (useful for plotting)
 $\dot{x}$: linear velocity of base

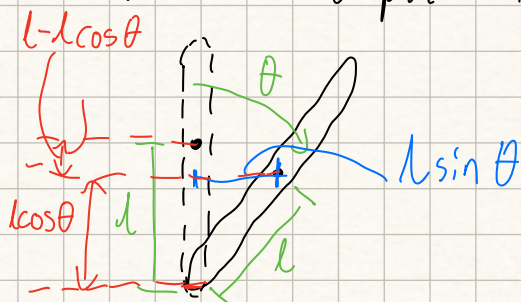
Model



Horizontal displacement of cart

$$M \frac{d^2 x}{dt^2} = u - H$$

Horizontal displacement of pendulum

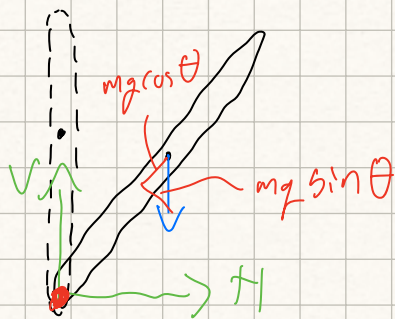


$$m \frac{d^2}{dt^2} (x + l \sin \theta) = H$$

Vertical displacement of pendulum

$$m \frac{d^2}{dt^2} (-l \cos \theta) = mg - H$$

Angular displacement of pendulum



$$J \frac{d^2 \theta}{dt^2} = l m g \sin \theta$$

$$M \frac{d^2 x}{dt^2} = a - H$$

$$m \frac{d^2}{dt^2} (x + l \sin \theta) = H$$

$$m \frac{d^2}{dt^2} (-l \cos \theta) = mg - V$$

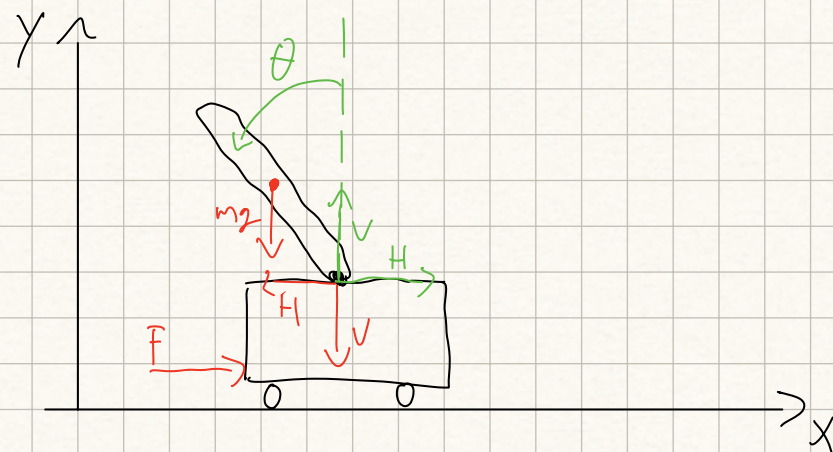
$$J \frac{d^2 \theta}{dt^2} = l m g \sin \theta$$

For small angles: $\cos \theta \approx 1$, $\sin \theta \approx \theta$

$$m \frac{d^2}{dt^2} (x + l \sin \theta) = H \Rightarrow m \frac{d^2}{dt^2} (x + l \theta) = H$$

$$m \frac{d^2}{dt^2} (-l \cos \theta) = mg - V \Rightarrow 0 = mg - V$$

$$J \frac{d^2 \theta}{dt^2} = l m g \sin \theta \Rightarrow J \frac{d^2 \theta}{dt^2} = l m g \theta$$



Horizontal forces on cart:

$$M \ddot{x} = F - H$$

Horizontal forces on pendulum:

$$m \frac{d^2}{dt^2} (x - l \sin \theta) = H$$

$$m \frac{d}{dt} \left[\frac{d}{dt} (x - l \sin \theta) \right] = H$$

$$m \frac{d}{dt} \left[\frac{dx}{dt} - \frac{d}{dt} l \sin \theta \right] = H$$

$$m \frac{d}{dt} \left[\dot{x} - l \cos \theta \cdot \dot{\theta} \right] = H$$

$$m \left(\frac{d\dot{x}}{dt} - \frac{d}{dt} l \cos \theta \cdot \dot{\theta} \right) = H$$

$$m \left(\ddot{x} - \left[\left(\frac{d}{dt} l \cos \theta \right) \cdot \dot{\theta} + l \cos \theta \cdot \frac{d}{dt} \dot{\theta} \right] \right) = H$$

$$m(\ddot{x} - [-l \sin \theta \cdot \dot{\theta} \cdot \dot{\theta} + l \cos \theta \cdot \ddot{\theta}]) = H$$

$$m(\ddot{x} + l \sin \theta \dot{\theta}^2 - l \cos \theta \ddot{\theta}) = H$$

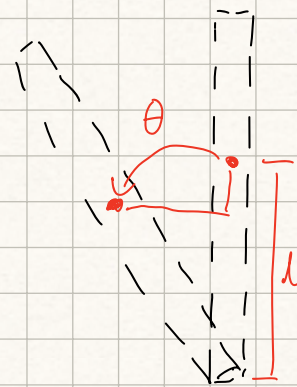
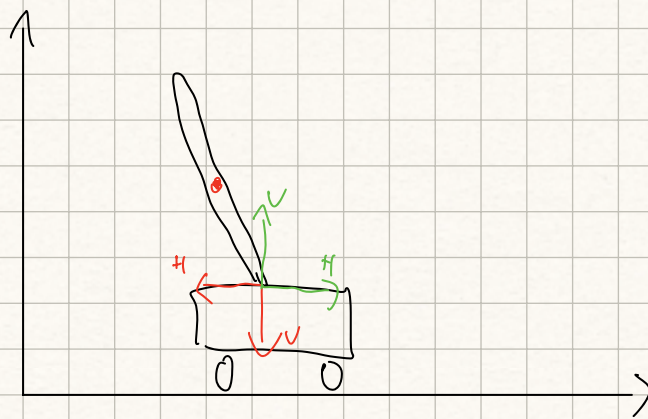
Eliminate H by putting the second into first:

$$M\ddot{x} = F - [m(\ddot{x} + l \sin \theta \dot{\theta}^2 - l \cos \theta \ddot{\theta})]$$

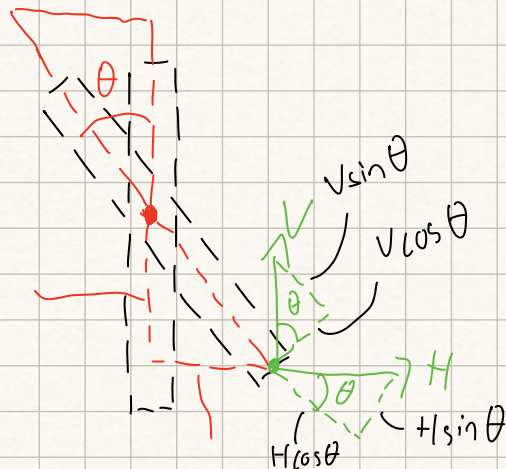
$$F - M\ddot{x} - m\ddot{x} - ml \sin \theta \dot{\theta}^2 + ml \cos \theta \ddot{\theta} = 0$$

$$F - M\ddot{x} - m\ddot{x} - ml(\sin \theta \dot{\theta}^2 - \cos \theta \ddot{\theta}) = 0$$

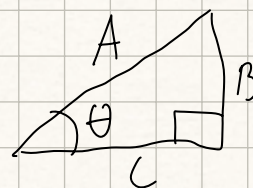
Forces and moments acting on pole:

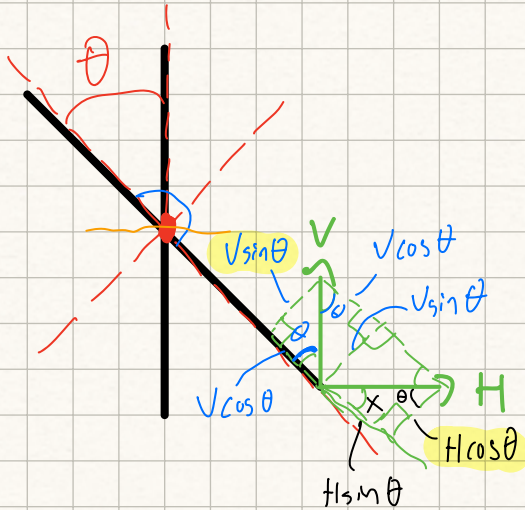


let the center of rotation be at the COM:



$$J\ddot{\theta} =$$





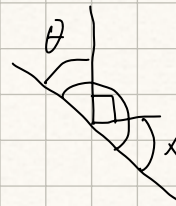
$$\sin \theta = \frac{B}{A} \Rightarrow B = A \sin \theta$$

$$\cos \theta = \frac{C}{A} \Rightarrow C = A \cos \theta$$

$$180 = 90 + \theta + x$$

$$\theta + x = 90$$

$$x = 90 - \theta$$



$$180 = 90 + x + y$$

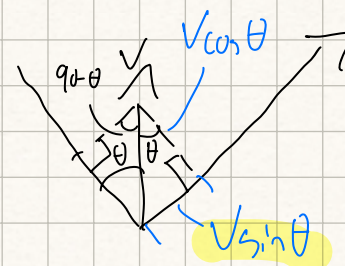
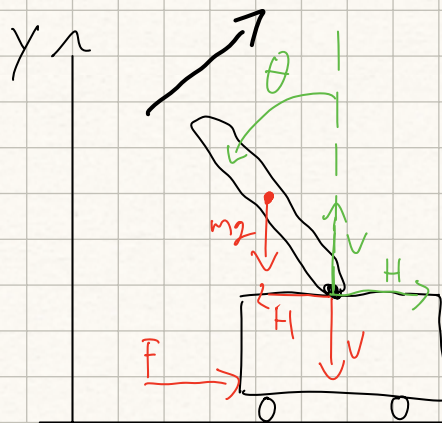
$$90 = x + y$$

$$90 = 90 - \theta + y$$

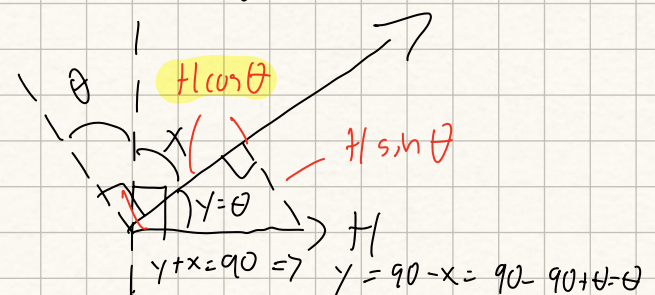
$$\theta = y$$

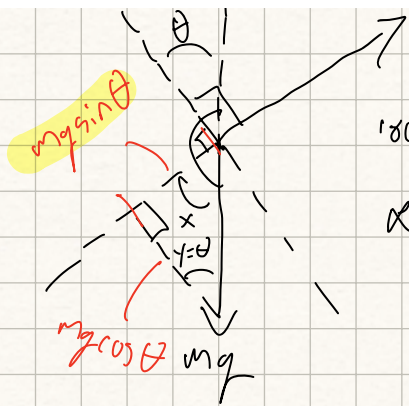
$$J \ddot{\theta} = L V \sin \theta + L H \cos \theta$$

Summarize the forces perpendicular to the pendulum:



$$x + \theta = 90 \Rightarrow x = 90 - \theta$$

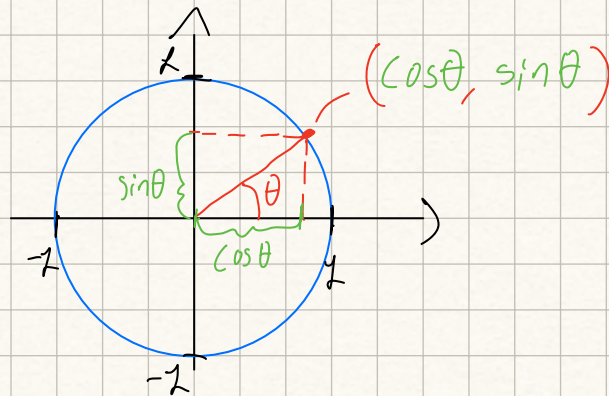
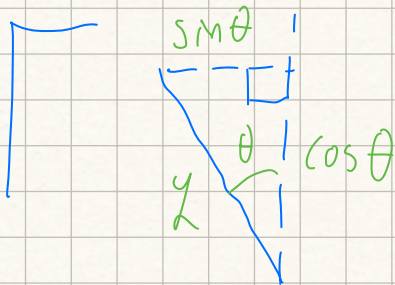




$$180 = 90 + \theta + x \Rightarrow x = 90 - \theta$$

$$x + y = 90 \Rightarrow y = 90 - x = 90 - 90 + \theta = \theta$$

$$V \sin \theta + T \cos \theta - mg \sin \theta =$$



$$F = ma, \quad F = V \sin \theta + T \cos \theta - mg \sin \theta$$

$$a = ?$$

Acceleration of pendulum COM along the perpendicular axis

$$\frac{d^2}{dt^2} (\text{position})$$

$$\text{position: } \begin{bmatrix} x - l \sin \theta \\ l \cos \theta \end{bmatrix}$$

$$\frac{d^2}{dt^2} \begin{bmatrix} x - l \sin \theta \\ l \cos \theta \end{bmatrix} = \frac{d}{dt} \begin{bmatrix} \dot{x} - l \cos \theta \cdot \dot{\theta} \\ -l \sin \theta \cdot \dot{\theta} \end{bmatrix} = \begin{bmatrix} \ddot{x} + l \sin \theta \cdot \dot{\theta} \cdot \dot{\theta} - l \cos \theta \cdot \ddot{\theta} \\ -l \cos \theta \cdot \dot{\theta} \cdot \dot{\theta} - l \sin \theta \cdot \ddot{\theta} \end{bmatrix}$$

$$= \begin{bmatrix} \ddot{x} + l \sin \theta \cdot \ddot{\theta}^2 - l \cos \theta \cdot \ddot{\theta} \\ -l \cos \theta \cdot \ddot{\theta}^2 - l \sin \theta \cdot \ddot{\theta} \end{bmatrix} = \ddot{\mathbf{r}}$$

Now project onto perpendicular axis $\begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} = \hat{\mathbf{e}}_\theta$

$$a_\theta = \ddot{\mathbf{r}} \cdot \hat{\mathbf{e}}_\theta$$

$$a_\theta = \begin{bmatrix} \ddot{x} + l \sin \theta \cdot \ddot{\theta}^2 - l \cos \theta \cdot \ddot{\theta} \\ -l \cos \theta \cdot \ddot{\theta}^2 - l \sin \theta \cdot \ddot{\theta} \end{bmatrix} \cdot \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}$$

$$= \ddot{x} \cos \theta + \cancel{l \ddot{\theta}^2 \sin \theta \cos \theta} - l \ddot{\theta} \cos^2 \theta - \cancel{l \ddot{\theta}^2 \sin \theta \cos \theta} - l \ddot{\theta} \sin^2 \theta$$

$$= \ddot{x} \cos \theta - l \ddot{\theta} (\sin^2 \theta + \cos^2 \theta)$$

$$= \ddot{x} \cos \theta - l \ddot{\theta}$$

$$F = ma_i$$

$$V \sin \theta + T \cos \theta - mg \sin \theta = m \ddot{x} \cos \theta - ml \ddot{\theta} \quad (2)$$

$$J \ddot{\theta} = l V \sin \theta + l T \cos \theta \quad (1)$$

Need to eliminate V and T :

$$l V \sin \theta = J \ddot{\theta} - l T \cos \theta \Rightarrow V = \frac{J \ddot{\theta}}{l \sin \theta} - \frac{T \cos \theta}{\sin \theta}$$

Putting this into (2):

$$\left(\frac{J \ddot{\theta}}{l \sin \theta} - \frac{T \cos \theta}{\sin \theta} \right) \sin \theta + T \cos \theta - mg \sin \theta = m \ddot{x} \cos \theta - ml \ddot{\theta}$$

$$\frac{J\ddot{\theta}}{L} - l\cos\theta + l\cos\theta - mg\sin\theta = m\ddot{x}\cos\theta - ml\ddot{\theta}$$

$$\frac{J\ddot{\theta}}{L} - mg\sin\theta - m\ddot{x}\cos\theta + ml\ddot{\theta} = 0$$

$$J\ddot{\theta} + ml^2\ddot{\theta} - ml\ddot{x}\cos\theta - mlgsin\theta = 0$$

The governing equations:

$$F - M\ddot{x} - m\ddot{x} - ml(\sin\theta\dot{\theta}^2 - \cos\theta\ddot{\theta}) = 0$$

$$J\ddot{\theta} + ml^2\ddot{\theta} - ml\ddot{x}\cos\theta - mlgsin\theta = 0$$

$$\ddot{x} = \frac{F - ml(\sin\theta\dot{\theta}^2 - \cos\theta\ddot{\theta})}{(M+m)}$$

$$\ddot{\theta} = \frac{ml\ddot{x}\cos\theta + mlgsin\theta}{J + ml^2} = \frac{ml(\ddot{x}\cos\theta + gsin\theta)}{J + ml^2}$$

Linearizing

position does not matter for our system, so we look at small perturbations to angle

Assume angle will stay small, around $\theta = 0$

Let ϕ represent a small deviation from the equilibrium:

$$\cos(0 + \phi) \approx 1$$

$$\sin(0 + \phi) \approx \phi$$

$$\ddot{\theta} = \ddot{\phi} \approx 0$$

The linearized equations become;

$$F - M\ddot{x} - m\ddot{x} - ml(\sin\theta \ddot{\theta}^2 - \cos\theta \ddot{\theta}) = 0$$

$$\Rightarrow F - (M+m)\ddot{x} + ml\ddot{\theta} = 0$$

$$J\ddot{\theta} + ml^2\ddot{\theta} - ml\ddot{x}\cos\theta - mlg\sin\theta = 0$$

$$\Rightarrow (J + ml^2)\ddot{\theta} - ml\ddot{x} - mlg\phi = 0$$

State-Space Representation

$$\mathbf{z} = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \end{bmatrix} = \begin{bmatrix} \theta \\ \dot{\theta} \\ x \\ \dot{x} \end{bmatrix}$$

$$\dot{\mathbf{z}} = \begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \\ \dot{z}_4 \end{bmatrix} = \begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \\ \dot{x} \\ \ddot{x} \end{bmatrix} = \begin{bmatrix} z_2 \\ \frac{ml(\dot{z}_4 \cos z_1 + g \sin z_1)}{J + ml^2} \\ z_4 \\ \frac{F - ml(z_2^2 \sin z_1 - \dot{z}_2 \cos z_1)}{M+m} \end{bmatrix}$$

Fully expanded:

$$= \begin{bmatrix} Z_2 \\ \frac{ml \ddot{Z}_4 \cos Z_1 + ml g \sin Z_1}{J + ml^2} \\ Z_4 \\ \frac{F - ml \dot{Z}_2^2 \sin Z_1 + ml \ddot{Z}_2 \cos Z_1}{M + m} \end{bmatrix}$$